A Fast Array RLS Equalizer for Phase Coherent Underwater Acoustic Communication

Fei-Yun Wu, *Senior Member, IEEE*, Hui-Zhong Yang, and Zhenwen Liu

***Abstract*—To overcome the $\mathcal{O}(L^2)$ complexity of the recursive least squares (RLS) algorithm, this article proposes a more computationally efficient fast array adaptive equalizer (FAE). Benefiting from the array representation and structural adjustments of the equalizer input vector, the proposed fast array RLS (FARLS) algorithm with the linear $\mathcal{O}(L)$ complexity is derived while ensuring performance. The comprehensive comparison of current adaptive algorithms is conducted, including the classical RLS and least mean square algorithms, as well as the emerging proportionate RLS and data-reusing RLS algorithms. Through two aspects of validation, including computational efficiency and steady-state mean square deviation, the FARLS algorithm is successfully demonstrated to have advantages over the above algorithms. In addition, we combine the FARLS algorithm with phase coherent in FAE to further enhance equalization performance, and validate its noise robustness and Doppler shift resistance through simulations and at-sea experiments. Real data shows that FAE outperforms its counterparts by at least 42.1% in terms of the percentage the percentage of frames with BER = 0.**

***Index Terms*—Adaptive equalizer, fast array recursive least squares (FARLS), phase tracking, underwater acoustic (UWA) communication.**

I. Introduction

UNDERWATER acoustic (UWA) channel is a delay Doppler doubly spread channel [1], [2], posing significant challenges to reliable communication due to time-varying multipath propagation and Doppler effects. The most critical problems include long delay spread, severe intersymbol interference (ISI), and time-varying frequency offsets, all of which hinder efficient signal recovery and demodulation. The challenges posed by delay Doppler doubly spread channel have significantly impacted the UWA communication. The primary threats arise from multipath effects and Doppler shift [4]. Multipath causes time delay expansion in the transmitted information [5], much larger than that of terrestrial wireless communication [6], [7]. Doppler shift results in a frequency offset of the transmitted signal, whereas in radio-frequency communication, Doppler shift is often ignored [8]. High propagation loss limits the bandwidth of UWA channel. These characteristics necessitate robust channel equalization techniques to ensure reliable underwater information transmission.

In order to address the above problems and enhance the reliability and efficiency of underwater information transmission, researchers have extensively explored various methods, such as channel estimation and channel equalization [9], [10], [11].

Orthogonal frequency division multiplexing (OFDM), frequency-shift keying (FSK), and phase-shift keying (PSK) are widely used in UWA communication systems [12], [13]. OFDM improves bandwidth utilization by transmitting multiple subcarriers in parallel, but its orthogonality is susceptible to frequency offsets [14]. FSK modulation has less sensitivity to frequency variations and is less susceptible to interference, making it suitable for anti-Doppler UWA communication. PSK modulation is commonly used in coherent communication for high-speed information transmission [15]. M-ary PSK (MPSK)-based virtual time-reversal-mirror is used in multisequence superimposed spread spectrum, to get a lower bit error rate (BER) in UWA communication [16]. Furthermore, experiments indicate that the combination of MPSK signals with decision feedback equalizer can achieve the data rate of 200 bits/s in UWA communication [17].

To enhance the efficiency and reliability of UWA communication, researchers have extensively explored many techniques based on the least mean square (LMS) and recursive least squares (RLS) algorithms. To improve the accuracy of the LMS algorithm, the normalized version (NLMS) algorithm, proportionate NLMS (PNLMS) algorithm, and improved PNLMS (IPNLMS) algorithm have been designed [19], [20], [21]. In echo cancellation experiments, the IPNLMS algorithm demonstrates superior performance compared to NLMS and PNLMS algorithms. The norm strategy is also introduced into the cost function, such as integrating the $\ell_{1,0}$-norm constraint in the IPNLMS algorithm, referred to as the INPMLS-BL0 algorithm. Experimental results demonstrate a significant advantage of INPMLS-BL0 over the IPNLMS algorithm in terms of noise robustness and tracking performance [22]. Similarly, an IPNLMS algorithm based on the mixing regularization parameter (MRP-IPNLMS) has been developed. In the multiple-input multiple-output UWA experiment with six receivers, the channel estimation error of the MRP-IPNLMS algorithm is reduced by 2.5 dB [23]. In addition,

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designing the step size of the LMS algorithm is another method to enhance convergence speed [24], [25].

RLS algorithms that use the sparse structure of the system have been studied, including both proportional RLS (PRLS) and block sparse RLS (BSRLS) algorithms [26], [27], [28]. The data-reusing RLS (DR-RLS) algorithm has also enhanced the accuracy of the RLS algorithm, especially in terms of tracking capability [29]. The PRLS, BSRLS, and DR-RLS algorithms have respectively enhanced the accuracy of RLS algorithm in sparse and general scenarios. However, traditional RLS algorithms are computationally intensive and numerically unstable, especially in dynamic underwater environments. Array algorithms, as the effective alternative to RLS algorithm, have given rise to the family of orthogonal–triangular (QR)-decomposition-based algorithms [30], [31]. Inspired by the array structure with more stable numerical properties, we recognize the necessity of designing faster versions to accommodate the resource-constrained environment of underwater communication.

Methods for reducing the complexity of RLS are achieved through order-recursive updates of the design parameters, such as in lattice algorithms [32], [33], or by accelerating the time updates with the fixed-order through changes in data structures, such as in transversal filters [34], [35]. We are more interested in the latter, as the efficient transformation of data structures is intricately coupled with array theory. This article derives the fast array RLS (FARLS) algorithm presented in the array form. The FARLS algorithm exhibits more stable numerical characteristics, and its computational complexity is reduced to $\mathcal{O}(L)$.

To address the inefficiency and instability of conventional RLS algorithms in UWA scenarios, this article proposes FARLS adaptive equalizer with a phase-coherent mechanism in FAE. The proposed solution achieves robust and low-complexity equalization, making it well-suited for dynamic UWA environments.

The main contributions to this article are as follows.

1) We propose the fast array adaptive equalizer (FAE) for UWA communication and design the phase coherent mechanism. The FAE has excellent noise robustness and resistance to Doppler shift, and its ability to quickly compensate for rapidly rotating channel taps enhances equalization performance in dynamic environment.
2) Furthermore, by introducing array form expressions and transforming the structure of the equalizer input vector, we derive the FARLS algorithm that has the computational complexity of only $\mathcal{O}(L)$. This study also analyze the differences between the LMS, RLS, PRLS, DR-RLS, and FARLS algorithms, including the principles, operability, and computational complexity.
3) We design UWA communication simulations and at-sea experiments to validate the BER performance of the proposed FAE, and we have expanded on its potential development directions and application scenarios.

The rest of this article is organized as follows. In Section II, we provide a detailed description of the phase coherent UWA communication system model. In Section III, we discuss the derivation process of the FARLS algorithm and explained its joint working mechanism with phase coherent. In Section IV, we analyze the convergence of the FARLS algorithm and validated its BER performance in UWA simulations. In Section V, we conduct at-sea experiments to compare the equalization performance of the FAE and provided explanations on its future development directions. Finally Section VI, concludes this article.

Notations: L and N are the length of channel and transmitted signals. $(\cdot)^*$, $(\cdot)^T$, and $(\cdot)^H$ denotes the complex conjugate, transpose, and Hermitian transpose operators, respectively.

II. Framework

In this section, we presented the overall process of the phase-locked communication system and discussed the main challenges of underwater data transmission. A well-structured transmission frame, along with an optimized receiver, plays a crucial role in ensuring the efficient and reliable reconstruction of signals. The transmitted signal after PSK modulation at time n is

$$d(n) = \mathrm{Re}\left[\rho(n)e^{j(2\pi f_c n+\theta(m))}\right] \tag{1}$$

where $n = 1, 2, \ldots, N$ and $\rho(n)$ is the signal waveform. m denotes the possible phases as the carrier. $\theta_m = 2\pi(m-1)/M$, $m = 1, 2, \ldots, M$.

Fig. 1 is the transmitted data frame. Both up linear frequency modulation (up-LFM) and down-LFM signals are used for frame synchronization. The frequency of up-LFM increases with the increase in transmission frequency, while down-LFM exhibits the opposite frequency trend. The t_g, t_u, and t_d represent the duration of guard space, up-LFM signal, and down-LFM signal, respectively. The N_t and N_s denote the length of training and information symbols. In Sections IV-C and V, we use the phase coherent communication system shown in Fig. 2 for both simulations and at-sea experiments.

At time n, the received signal is

$$x(n) = e^{j\varphi(n)} \sum_{l=1}^{L} c_l(n) d\,(n-l) + v(n) \tag{2}$$

where $\varphi(n)$ presents the time-varying phase rotation, $c_l(n)$ is the lth channel tap, and $v(n)$ denotes the additive Gaussian white noise. It is a major challenge in underwater data transmission to mitigate the adverse effects of rapid variations of $\varphi(n)$. Therefore, designing an efficient and reliable adaptive equalizer is essential.

Fig. 3 shows the overall structure of FAE, whose advantages in low complexity and robustness are attributed to the effective combination of weight vector $\mathbf{w}(n)$ update methods and phase coherent. We achieve linear complexity updates through the FARLS algorithm, and the rapid estimation and compensation of $\varphi(n)$ improved the equalization performance. The output of the FAE is

$$\begin{aligned}\hat{d}(n) &= \mathbf{w}^H(n)\mathbf{s}(n) \\ &= \left[\mathbf{w}_1^H(n), -\mathbf{w}_2^H(n)\right]\left[e^{-j\hat{\varphi}(n)}\mathbf{x}_1^H(n), \tilde{\mathbf{d}}^H(n)\right]^H\end{aligned} \tag{3}$$

where $\mathbf{w}_1(n)$ and $\mathbf{w}_2(n)$ are the feedforward and feedback weight vectors, respectively, with dimensions $L_1 \times 1$ and

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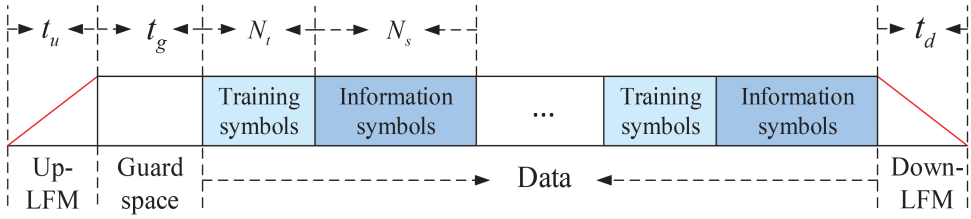


Fig. 1. Design of transmitted data for phase coherent UWA communication.

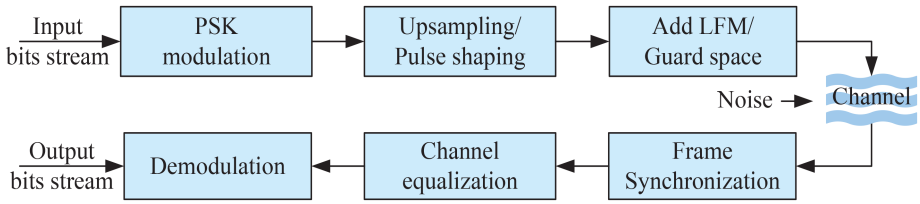


Fig. 2. Overall structure of phase coherent UWA communication system.

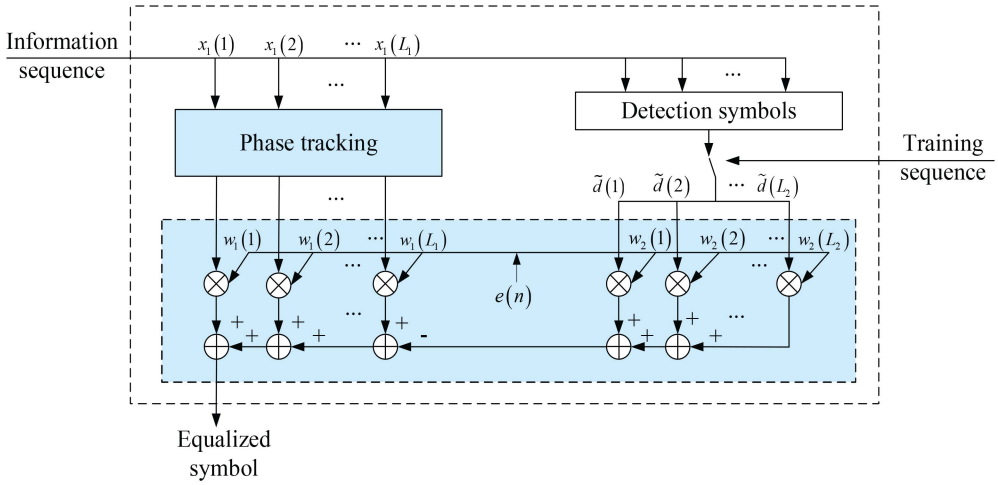


Fig. 3. Overall structure of the proposed FARLS adaptive equalizer.

$L_2 \times 1$, and $L = L_1 + L_2$. The relationship among $\mathbf{w}(n)$, $\mathbf{w}_1(n)$, and $\mathbf{w}_2(n)$ is $\mathbf{w}(n) = [\mathbf{w}_1^H(n), \mathbf{w}_2^H(n)]^H$. $\mathbf{s}(n)$ is a complex vector and $\hat{\varphi}(n)$ represents the estimated phase rotation. Applying the time window $[n - L_1 + 1, n]$ to $\mathbf{x}(n)$ yields $\mathbf{x}_1(n)$ with dimensions $L_1 \times 1$, where

$$\mathbf{x}_1(n) = [x(n), x(n-1), \ldots, x(n - L_1 + 1)]^H.$$

$\tilde{d}(n)$ is the previously detected symbols, with dimensions $L_2 \times 1$. As shown in Fig. 3, during the training process of the FAE, the training symbols are directly used as $\tilde{\mathbf{d}}(n)$.

The objective function of FAE

$$e(n) = d(n) - \hat{d}(n) \tag{4}$$

where $e(n)$ is iteratively minimized in the FARLS algorithm to achieve the optimal $\mathbf{w}(n)$. In Section III-C, we provide the detailed description of how $\mathbf{w}(n)$ and $\hat{\varphi}(n)$ are jointly working in the proposed FAE.

III. Algorithm Design

In this section, Section III-A analyzes the traditional RLS algorithm and its variants. Next, for array algorithms, Section III-B provides the theoretical origins of array methods and reveals the connection between prearray and postarray. Finally, in Section III-C, the FARLS algorithm is designed and works jointly with phase coherent. Together with (3), we complete the theoretical derivation of the FAE.

A. Review and Comparison of Adaptive Filtering Algorithms

To facilitate clearer understanding and comparison, we systematically categorize the representative adaptive filtering algorithms LMS, NLMS, PNLMS, IPNLMS, RLS, PRLS, and DR-RLS based on their structure, application scope, convergence characteristics, and computational complexity.

Table I provides a structured summary of these algorithms. The LMS family (including its normalized and proportionate

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TABLE I
COMPARISON OF LMS, NLMS, PNLMS, IPNLMS, RLS, DR-RLS, AND PRLS ALGORITHMS IN PRINCIPLE, APPLICATION SCOPE, PERFORMANCE, AND COMPLEXITY

Algorithm	Principle	Application Scope	Performance Feature	Complexity
LMS	$\mathbf{w}(n)=\mathbf{w}(n-1)+p\mathbf{x}(n)e^*(n)$	General-purpose adaptive filtering	Simple implementation, sensitive to step size p	$\mathcal{O}(L)$
NLMS	$\mathbf{w}(n)=\mathbf{w}(n-1)+\frac{p}{\|\mathbf{x}(n)\|^2+\varepsilon}\mathbf{x}(n)e^*(n)$	Signal-dependent scenarios, better for time-varying power	Improved stability and convergence over LMS	$\mathcal{O}(L)$
PNLMS	$\mathbf{w}(n)=\mathbf{w}(n-1)+\mu\mathbf{G}(n)\mathbf{x}(n)e^*(n)$	Sparse systems	Improves convergence on sparse channels	$\mathcal{O}(L)$
IPNLMS	Mixed update rules combining uniform and proportionate adaptation	Both sparse and non sparse channels	Robust across diverse scenarios, balances adaptation	$\mathcal{O}(L)$
RLS	$\mathbf{w}(n)=\mathbf{w}_L(n-1)+\mathbf{k}(n)e^*(n)$	Fast convergence in dynamic channels	Strong performance, but high computational burden	$\mathcal{O}(L^2)$
PRLS	$\mathbf{w}(n)=\mathbf{w}(n-1)+\mathbf{G}(n-1)\mathbf{k}(n)e^*(n)$ $\mathbf{G}(n-1)=\mu\text{diag}\{g_L(n-1)\}$	Sparse dynamic systems	Incorporates sparsity awareness, faster convergence than RLS in sparse settings	$\mathcal{O}(L^2)$
DR-RLS	$\mathbf{w}(n)=\mathbf{w}(n-1)+\beta(n)\mathbf{k}(n)e^*(n)$ $\beta(n)=\frac{1-\alpha^c(n)}{1-\alpha(n)}$	Time-varying channels needing fast tracking	Higher precision with increased parameter c; reduces to RLS when c=1	$\mathcal{O}(L^2)$

variants) offers low-complexity solutions suitable for real-time and sparse environments, while the RLS family achieves faster convergence at the cost of higher computational load. NLMS introduces normalization to stabilize adaptation; PNLMS and IPNLMS exploit signal sparsity for improved performance. For fast-tracking dynamic channels, PRLS incorporates sparsity-awareness into RLS updates, and DR-RLS enhances RLS tracking capability through recursive gain control.

This comparison helps motivate our proposed FARLS algorithm, which aims to retain the fast convergence of RLS-type algorithms while significantly reducing computational complexity.

The classical RLS algorithm is dedicated to addressing the regularized least squares problem. At time n, the RLS algorithm is represented as follows:

$$\begin{aligned} e(n) &= d(n)-\mathbf{w}^H(n-1)\mathbf{x}(n)\\ \mathbf{k}(n) &= \frac{\mathbf{Q}(n-1)\mathbf{x}(n)}{\lambda+\mathbf{x}^H(n)\mathbf{Q}(n-1)\mathbf{x}(n)}\\ \mathbf{Q}(n-1) &= \lambda^{-1}[1-\mathbf{x}^H(n)\mathbf{k}(n)]\mathbf{Q}(n-1)\\ \mathbf{w}(n) &= \mathbf{w}_L(n-1)+\mathbf{k}(n)e^*(n) \end{aligned} \tag{5}$$

where $e(n)$ denotes the posteriori error, $d(n)$ and $\mathbf{x}(n)$ stand for the ideal signal and the filter input signal, where the former is a scalar, and the latter is $L\times 1$. The $\mathbf{k}(n)$ is the $L\times 1$ gain vector, $\mathbf{Q}(n)$ is the $L\times L$ autocorrelation matrix, and $\mathbf{w}(n)$ represents the filter coefficients, with dimensions $L\times 1$. λ serves as the forgetting factors and $0\ll\lambda<1$.

The purpose of the PRLS algorithm is to use the sparse structure in the channel to expedite the convergence of zero coefficients. On the other hand, the DR-RLS algorithm repeatedly uses a set of data multiple times, enhancing the stability of the tap coefficients. Algorithms 1 and 2 present detailed information about the PRLS algorithm and the DR-RLS algorithm. In Algorithm 1, $\mathbf{G}$ is the scaling matrix composed of a fixed step size μ and a variable step size $g_k(n-1)$, where ϵ is the regularization parameter, $-1\le\vartheta<1$, and $k=1,2,\ldots,L$. In Algorithm 2, the parameter c denotes the data reuse times within a single iteration.

B. Array Methods

The array algorithm is considered an equivalent variant of the RLS algorithm, but it is noteworthy that array methods provides improved numerical stability through norm preservation and inner product preservation. This stability arises from the array algorithm s use of prearrays and postarrays for updating entries, rather than the traditional explicit equations form. The transition from prearray to postarray typically requires triangulation operations. Assuming that the lower triangular matrix $\mathbf{C}$ and a general matrix $\mathbf{F}$ satisfy

$$\begin{aligned} \mathbf{CC}^H &= \mathbf{AA}^H+\mathbf{BSB}^H\\ \mathbf{FC}^H &= \mathbf{DA}^H+\mathbf{ESB}^H \end{aligned} \tag{6}$$

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Algorithm 1: PRLS Algorithm.

Inputs: Signal vector $\mathbf{x}(n)$, desired signal $d(n)$, forgetting factor λ, filter length L
Outputs: Estimated coefficient vector $\mathbf{w}(n)$
Initialization: Set $\mathbf{w}(0)=\mathbf{0}$, $\mathbf{P}(0)=\delta^{-1}\mathbf{I}$ `// Initial weight and inverse correlation matrix`
for $n=1,2,\ldots,N$
$e(n)=d(n)-\mathbf{w}^H(n-1)\mathbf{x}(n)$
$\mathbf{k}(n)=\mathbf{Q}(n-1)\mathbf{x}(n)/(\lambda+\mathbf{x}^H(n)\mathbf{Q}(n-1)\mathbf{x}(n))$ `// Gain vector update`
$\mathbf{G}(n-1)=\mu\text{diag}\{g_L(n-1)\}$
$g_k(n-1)=$
$(1+\vartheta)/2L+(1-\vartheta)|w_k(n-1)|/(2\|\mathbf{w}(n-1)\|_1+\varepsilon)$,
$\mathbf{Q}(n)=\lambda^{-1}[1-\mathbf{x}^H(n)\mathbf{k}(n)]\mathbf{Q}(n-1)$
$\mathbf{w}(n)=\mathbf{w}(n-1)+\mathbf{G}(n-1)\mathbf{k}(n)e^*(n)$ `// Coefficient update`
end

Algorithm 2: DR-RLS Algorithm.

Input: λ (forgetting factor), c (tracking parameter), $d(n)$ (desired signal), $\mathbf{x}(n)$ (input vector)
Output: $\mathbf{w}(n)$ (filter coefficients)
Initialization:
Set $\mathbf{Q}(0)=\mathbf{I}_L$, $\mathbf{w}(0)=\mathbf{k}(0)=\mathbf{0}_{L\times 1}$
for $n=1,2,\ldots,N$ **do**
$e(n)=d(n)-\mathbf{x}^H(n)\mathbf{w}_L(n)$ // Estimation error
$\mathbf{k}(n)=\mathbf{Q}(n-1)\mathbf{x}(n)/(\lambda+\mathbf{x}^H(n)\mathbf{Q}(n-1)\mathbf{x}(n))$ // Gain vector
$\alpha(n)=1-\mathbf{x}^H(n)\mathbf{k}(n)$ // Innovation factor
$\beta(n)=1-\alpha^c(n)/(1-\alpha(n))$ // Data-reusing factor
$\mathbf{Q}(n)=\lambda^{-1}[1-\mathbf{x}^H(n)\mathbf{k}(n)]\mathbf{Q}(n-1)$ // Update inverse correlation matrix
$\mathbf{w}(n)=\mathbf{w}(n-1)+\beta(n)\mathbf{k}(n)e^*(n)$ // Update filter coefficients
end for

where $\mathbf{A}$, $\mathbf{B}$, $\mathbf{D}$, and $\mathbf{E}$ are matrices or vectors. $\mathbf{T}$ is the extended matrix of the signature matrix $\mathbf{S}$ and

$$\mathbf{T}=\begin{bmatrix}\mathbf{I} & \\ & \mathbf{S}\end{bmatrix}.$$

We solve (6) using the array approach. First, we form a prearray as follows:

$$\mathbf{P}=\begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{D} & \mathbf{E}\end{bmatrix}. \tag{7}$$

The postarray is a lower triangular matrix, and its relationship with the prearray $\mathbf{P}$ is

$$\underbrace{\begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{D} & \mathbf{E}\end{bmatrix}}_{\mathbf{P}}\Theta=\underbrace{\begin{bmatrix}h & \mathbf{0}\\ \mathbf{y} & \mathbf{Z}\end{bmatrix}}_{\mathbf{R}} \tag{8}$$

where the unitary matrix Θ represents a sequence of Givens rotations and Householder transformations, which lower triangularize $\mathbf{P}$ into the postarray. Θ is also the unitary matrix of the $\mathbf{T}$, i.e., $\Theta\mathbf{T}\Theta^H=\mathbf{T}$. h, $\mathbf{y}$, and $\mathbf{Z}$ are lower triangular matrix with diagonal elements greater than 0, square matrix, and general matrix, respectively. Furthermore, squaring both sides of (8)

$$\begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{D} & \mathbf{E}\end{bmatrix}\Theta\begin{bmatrix}\mathbf{I} & \\ & \mathbf{S}\end{bmatrix}\Theta^H\begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{D} & \mathbf{E}\end{bmatrix}^H$$
$$=\begin{bmatrix}h & \mathbf{0}\\ \mathbf{y} & \mathbf{Z}\end{bmatrix}\begin{bmatrix}\mathbf{I} & \\ & \mathbf{S}\end{bmatrix}\begin{bmatrix}h & \mathbf{0}\\ \mathbf{y} & \mathbf{Z}\end{bmatrix}^H. \tag{9}$$

Expanding both sides of (9) in conjunction with the properties of $\mathbf{T}$ and comparing with (6), it is demonstrated that the h and $\mathbf{y}$ entries in the postarray correspond to $\mathbf{C}$ and $\mathbf{F}$.

C. Fast Array Equalizer

Based on the array methods, we will use the data structure to design an efficient RLS array algorithm. Assuming the filter input vector at n time is

$$\mathbf{x}(n)=[x(n),x(n-1),\ldots,x(n-L+2),x(n-L+1)]^T$$

and the $\mathbf{x}(n-1)$ is given by

$$\mathbf{x}(n-1)=[x(n-1),x(n-1),\ldots,x(n-L+1),x(n-L)]^T.$$

Both $\mathbf{x}(n)$ and $\mathbf{x}(n-1)$ contain numerous identical entries, leading to

$$\begin{bmatrix}x(n)\\ \mathbf{x}(n-1)\end{bmatrix}=\begin{bmatrix}\mathbf{x}(n)\\ x(n-L)\end{bmatrix}. \tag{10}$$

Using (10) is precisely the key to avoiding the inverse operation of $\mathbf{Q}(n)$, which is essential for achieving low linear complexity. It involves directly implementing the time update from $\mathbf{k}(n-1)$ to $\mathbf{k}(n)$. Fig. 4 shows the derivation process of the FARLS algorithm, where "Efficient shift structure" corresponds to the data structure described in (10). We provide the alternative expression for $\mathbf{k}(n)$ and $\mathbf{Q}(n)$

$$\mathbf{k}(n)=\lambda^{-1}\gamma(n)\mathbf{Q}(n-1)\mathbf{x}(n) \tag{11}$$

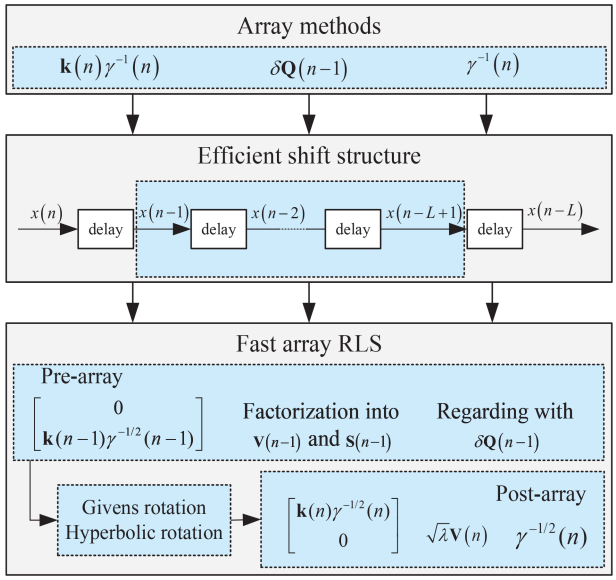


Fig. 4. Derivation process of the proposed FARLS algorithm.

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$$\mathbf{Q}(n) = \lambda^{-1}\mathbf{Q}(n-1) - \frac{\mathbf{k}(n)\mathbf{k}^H(n)}{\gamma(n)} \tag{12}$$

where $\gamma(n)$ is the conversion factor and we have

$$\gamma(n) = \frac{1}{1+\lambda^{-1}\mathbf{x}^H(n)\mathbf{Q}(n-1)\mathbf{x}(n)}. \tag{13}$$

We rewrite (11) as $\mathbf{k}(n)\gamma^{-1}(n) = \lambda^{-1}\mathbf{Q}(n-1)\mathbf{x}(n)$, and give $\mathbf{k}(n-1)\gamma^{-1}(n-1) = \lambda^{-1}\mathbf{Q}(n-2)\mathbf{x}(n-1)$. The right-hand side of these two equations involves $\mathbf{x}(n)$ and $\mathbf{x}(n-1)$, respectively, and we obtain the extended vectors

$$\begin{bmatrix}\mathbf{Q}(n-1)\mathbf{x}(n)\\ 0\end{bmatrix} = \begin{bmatrix}\mathbf{Q}(n-1) & \mathbf{0}_{L\times 1}\\ \mathbf{0}_{1\times L} & 0\end{bmatrix}\begin{bmatrix}x(n)\\ \mathbf{x}(n-1)\end{bmatrix} \tag{14}$$

$$\begin{bmatrix}0\\ \mathbf{Q}(n-2)\mathbf{x}(n-1)\end{bmatrix} = \begin{bmatrix}0 & \mathbf{0}_{1\times L}\\ \mathbf{0}_{L\times 1} & \mathbf{Q}(n-2)\end{bmatrix}\begin{bmatrix}x(n)\\ \mathbf{x}(n-1)\end{bmatrix}. \tag{15}$$

Using (14) and (15), we express the preliminary relationship between $\mathbf{k}(n)\gamma^{-1}(n)$ and $\mathbf{k}(n-1)\gamma^{-1}(n-1)$ as

$$\begin{bmatrix}\mathbf{k}(n)\gamma^{-1}(n)\\ 0\end{bmatrix} = \begin{bmatrix}0\\ \mathbf{k}(n-1)\gamma^{-1}(n-1)\end{bmatrix} + \lambda^{-1}\delta\mathbf{Q}(n-1)\begin{bmatrix}x(n)\\ \mathbf{x}(n-1)\end{bmatrix} \tag{16}$$

where the

$$\delta\mathbf{Q}(n-1) \triangleq \begin{bmatrix}\mathbf{Q}(n-1) & \mathbf{0}_{L\times 1}\\ \mathbf{0}_{1\times L} & 0\end{bmatrix} - \begin{bmatrix}0 & \mathbf{0}_{1\times L}\\ \mathbf{0}_{L\times 1} & \mathbf{Q}(n-2)\end{bmatrix} \tag{17}$$

and the initial value of $\mathbf{Q}(n-1)$ is

$$\mathbf{Q}(0) = \Pi^{-1} = \eta\cdot\mathrm{diag}\{\lambda^2, \lambda^3, \ldots, \lambda^{L+1}\}$$

where η denotes the positive scalar.

We focus on updating the $\gamma^{-1}(n)$ and provide

$$\gamma^{-1}(n-1) = 1 + \lambda^{-1}\mathbf{x}^H(n-1)\mathbf{Q}(n-2)\mathbf{x}(n-1)$$

based on the (13). Subtracting $\gamma^{-1}(n-1)$ from $\gamma^{-1}(n)$, and incorporating the definition of $\delta\mathbf{Q}(n-1)$ in (17), we obtain

$$\gamma^{-1}(n) = \gamma^{-1}(n-1) + \lambda^{-1} \begin{bmatrix}x^*(n) & \mathbf{x}^H(n-1)\end{bmatrix}\delta\mathbf{Q}(n-1)\begin{bmatrix}x(n)\\ \mathbf{x}(n-1)\end{bmatrix}. \tag{18}$$

To solve (16) and (18) in the forms of (6), we need to consider modifying the representation of $\delta\mathbf{Q}(n-1)$. The initial value of $\delta\mathbf{Q}(n-1)$ is expressed as $\delta\mathbf{Q}(0) = \eta\lambda^2\cdot\mathrm{diag}\{1, 0, \ldots, 0, -\lambda^L\}$. It is evident that this is a second-rank matrix and is factorized into a symmetric form

$$\delta\mathbf{Q}(0) = \lambda\cdot\mathbf{V}^H(0)\mathbf{S}(0)\mathbf{V}(0) \tag{19}$$

where

$$\mathbf{V}(0) = \underbrace{\begin{bmatrix}\sqrt{\eta\lambda} & 0 & \ldots & 0 & 0\\ 0 & 0 & \ldots & 0 & \sqrt{\eta\lambda^{L+1}}\end{bmatrix}}_{L+1}$$

and

$$\mathbf{S}(0) = \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}.$$

Furthermore, as referenced in [36], the $\mathbf{S}$ remains unchanged over time, and the rank of $\delta\mathbf{Q}(n)$ is always 2. We obtain the general form of (19) given by

$$\delta\mathbf{Q}(n-1) = \lambda\cdot\mathbf{V}^H(n-1)\mathbf{S}(n-1)\mathbf{V}(n-1). \tag{20}$$

Substitute (21) into (17) and (19), and express the resulting equations in the form of (6), we have

$$\gamma^{-3/2}(n) = \gamma^{-1}(n-1) + \begin{bmatrix}x^*(n) & \mathbf{x}^H(n-1)\end{bmatrix}\mathbf{V}^H(n-1)\mathbf{S}(n-1) \times \mathbf{V}(n-1)\begin{bmatrix}x(n)\\ \mathbf{x}(n-1)\end{bmatrix} \tag{21}$$

$$\begin{bmatrix}\mathbf{k}(n)\gamma^{-1/2}(n)\\ 0\end{bmatrix}\gamma^{-1/2}(n) = \begin{bmatrix}0\\ \mathbf{k}(n-1)\gamma^{-1/2}(n-1)\end{bmatrix}\gamma^{-1/2}(n-1) + \mathbf{V}^H(n-1)\mathbf{S}(n-1)\mathbf{V}(n-1)\begin{bmatrix}x(n)\\ \mathbf{x}(n-1)\end{bmatrix} \tag{22}$$

and

$$\mathbf{T} = \begin{bmatrix}1 & \\ & \mathbf{S}(n-1)\end{bmatrix}.$$

From (21), we extract elements to form left-hand side of (8), which is represented as follows:

$$\begin{bmatrix}\gamma^{-1/2}(n) & \begin{bmatrix}x^*(n) & \mathbf{x}^H(n-1)\end{bmatrix}\mathbf{V}^H(n-1)\\ \begin{bmatrix}0\\ \mathbf{k}(n-1)\gamma^{-1/2}(n-1)\end{bmatrix} & \mathbf{V}^H(n-1)\end{bmatrix}$$

$$\Theta(n) = \begin{bmatrix}h & \mathbf{0}\\ \mathbf{y} & \mathbf{Z}\end{bmatrix} \tag{23}$$

where $\Theta(n)$ is not a constant matrix. The formation process of $\Theta(n)$ is typically described using Givens transformation and Hyperbolic transformation. The specific transformation processes are demonstrated in Algorithms 4 and 5, respectively.

Simultaneously squaring both sides of (23) to obtain a form similar to (9), and using the property $\Theta(n)\mathbf{T}\Theta^H(n) = \mathbf{T}$, we finally derive the equations for solving h, $\mathbf{y}$, and $\mathbf{Z}$, as (24) shown at the bottom of the next page.

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According to (16), (18), and the definition of $\delta\mathbf{Q}(n-1)$ in (21), h and $\mathbf{y}$ in the first and second equations of the system (24) are

$$\begin{cases} h = \gamma^{-1/2}(n) \\ \mathbf{y} = \begin{bmatrix} \mathbf{k}(n)\gamma^{-1/2}(n) \\ 0 \end{bmatrix} \end{cases}. \tag{25}$$

Substituting (25) into the left side of the last equation in system (24) and using (12) and (21), we obtain

$$\begin{aligned} \mathbf{ZS}(n-1)\mathbf{Z}^H &= \delta\mathbf{Q}(n) \\ &= \lambda\mathbf{V}^H(n-1)\mathbf{S}(n-1)\mathbf{V}(n-1). \end{aligned} \tag{26}$$

As $\mathbf{S}$ remains unchanged during the iterations, the derivation of the $\mathbf{Z}$ matrix is

$$\mathbf{Z} = \sqrt{\lambda}\mathbf{V}^H(n). \tag{27}$$

Based on the results from (25) and (27), the postarray is written as

$$\begin{bmatrix} \gamma^{-1/2}(n) & \mathbf{0}_{1\times 2} \\ \begin{bmatrix} \mathbf{k}(n)\gamma^{-1/2}(n) \\ 0 \end{bmatrix} & \sqrt{\lambda}\mathbf{V}^H(n) \end{bmatrix}.$$

Extracting the elements $\mathbf{k}(n)\gamma^{-1/2}(n)$ and $\gamma^{-1/2}(n)$ from the postarray, the update equation for the $\mathbf{w}(n)$ is

$$\begin{aligned} \mathbf{w}(n) = \mathbf{w}(n-1) &+ \left[\mathbf{k}(n)\gamma^{-1/2}(n)\right]\left[\gamma^{1/2}(n)\right] \\ &\times \left[d(n) - \mathbf{x}^H(n)\mathbf{w}(n-1)\right]. \end{aligned} \tag{28}$$

The update of $\mathbf{w}(n)$ as shown in (28) accelerates the computation speed of FAE. In addition, a significant method to further enhance the reliability of FAE is the estimation of $\varphi(n)$. The phase coherent process requires the participation of $\mathbf{w}(n)$ and the use of the relationship we provided in (3), i.e., $\mathbf{w}(n) = [\mathbf{w}_1^H(n), \mathbf{w}_2^H(n)]^H$, which is also beneficial for obtaining $\hat{\varphi}(n)$. The phase coherent process is expressed as

$$\hat{\varphi}(n) = \hat{\varphi}(n-1) + K_{f1}\zeta(n-1) + K_{f2}\sum_{i=1}^{n-1}\zeta(n) \tag{29}$$

$$\zeta(n) = \mathrm{Im}\left\{\mathbf{w}_1^H(n)\mathbf{x}_1(n)e^{-j\hat{\varphi}(n)}\left[d(n) - \mathbf{w}_2^H(n)\tilde{\mathbf{d}}(n)\right]^*\right\} \tag{30}$$

where K_{f1} and K_{f2} are the proportional and integral tracking constants and $K_{f1} = 10K_{f2}$, respectively. According to the definition in (29), we obtain $\hat{\varphi}(n+1)$, and by subtracting $\hat{\varphi}(n)$ from $\hat{\varphi}(n+1)$, we have

$$\begin{aligned} \hat{\varphi}(n+1) = 2\hat{\varphi}(n) &- \hat{\varphi}(n-1) \\ &+ 11K_{f_2}\zeta(n) - 10K_{f_2}\zeta(n-1) \end{aligned} \tag{31}$$

further enhancing computational efficiency of the proposed FAE. Algorithm 3 provides the detailed explanation of the computational steps of FAE. Algorithms 4 and 5 are the methods we mentioned in Section III-B for triangulating the prearray, specifically Givens rotation and Hyperbolic rotation.

Algorithm 3: FARLS Adaptive Equalizer.

Input: forgetting factor λ, scaling parameter η, phase tracking gains K_{f1}, K_{f2}, desired signal $d(n)$, input vector $\mathbf{x}(n)$
Output: weight vector $\mathbf{w}(n)$
Initialization
1. Set $\mathbf{w}(0) = \mathbf{0}_{L\times 1}, \mathbf{k}(0) = \mathbf{0}_{L\times 1}, \mathbf{Q}(0) = \mathbf{\Pi}^{-1}, \gamma^{-1/2}(0) = 1$.
2. Set $\mathbf{V}(0) = \begin{bmatrix} \sqrt{\eta\lambda} & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & \sqrt{\eta\lambda^{L+1}} \end{bmatrix}$,
$\mathbf{S} = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}, \mathbf{T} = \begin{bmatrix} 1 & \\ & \mathbf{S} \end{bmatrix}$.
for $n = 1, 2, \dots, N$
// Construct the augmented matrix for rotations
$\mathbf{P} = \begin{bmatrix} \gamma^{-1/2}(n) & \begin{bmatrix} x(n) & \mathbf{x}^H(n-1) \end{bmatrix}\mathbf{V}^H(n-1) \\ \begin{bmatrix} 0 \\ \mathbf{k}(n-1)\gamma^{-1/2}(n-1) \end{bmatrix} & \mathbf{V}^H(n-1) \end{bmatrix}$
// Apply Givens rotation
$\Theta_1(n) = \mathrm{GivensRot}(\mathbf{P}(1,1), \mathbf{P}(1,2))$
$\mathbf{P}_1(n) = \mathbf{P}(n)\Theta_1(n)$
// Apply hyperbolic rotation for regularization
$\Theta_2(n) = \mathrm{HyperbolicRot}(\mathbf{P}_1(1,1), \mathbf{P}_1(1,3))$
$\mathbf{R} = \mathbf{P}_1(n)\Theta_2(n) = \begin{bmatrix} \gamma^{-1/2}(n) & \mathbf{0}_{1\times 2} \\ \begin{bmatrix} \mathbf{k}(n)\gamma^{-1/2}(n) \\ 0 \end{bmatrix} & \sqrt{\lambda}\mathbf{V}^H(n) \end{bmatrix}$
// Update weight vector
$\mathbf{w}(n) = \mathbf{w}(n-1) + \left[\mathbf{k}(n)\gamma^{-1/2}(n)\right]\left[\gamma^{-1/2}(n)\right]^{-1}[d(n) - \mathbf{x}^H(n)\mathbf{w}(n-1)]$
// Phase tracking error based on quadrature correlation
$\zeta(n) = \mathrm{Im}\left\{\mathbf{w}_1^H(n)\mathbf{x}_1(n)e^{-j\hat{\varphi}(n)}\left(d(n) - \mathbf{w}_2^H(n)\tilde{\mathbf{d}}(n)\right)^*\right\}$.
// Update phase estimate
$\hat{\varphi}_{(n+1)} = 2\hat{\varphi}(n) - \hat{\varphi}(n-1) + 11K_{f_2}\zeta(n) - 10K_{f_2}\zeta(n-1)$,
end

IV. Simulations

In this section, we design two tests to compare the computational complexity and learning curve of the FARLS algorithm

$$\begin{cases} |h|^2 = \gamma^{-1}(n-1) + \begin{bmatrix} x^*(n) & \mathbf{x}^H(n-1) \end{bmatrix}\mathbf{V}^H(n-1)\mathbf{S}(n-1)\mathbf{V}(n-1)\begin{bmatrix} x(n) \\ \mathbf{x}(n-1) \end{bmatrix} \\ \mathbf{y}h^* = \begin{bmatrix} 0 \\ \mathbf{k}(n-1)\gamma^{-1}(n-1) \end{bmatrix} + \mathbf{V}^H(n-1)\mathbf{S}(n-1)\mathbf{V}(n-1)\begin{bmatrix} x(n) \\ \mathbf{x}(n-1) \end{bmatrix} \\ \mathbf{yy}^H + \mathbf{ZS}(n-1)\mathbf{Z}^H = \begin{bmatrix} 0 \\ \mathbf{k}(n-1)\gamma^{-1/2}(n) \end{bmatrix}\begin{bmatrix} 0 \\ \mathbf{k}(n-1)\gamma^{-1/2}(n-1) \end{bmatrix}^H + \mathbf{V}(n-1)^H\mathbf{S}(n-1)\mathbf{V}(n-1). \end{cases} \tag{24}$$

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Algorithm 4 GivensRot.

Input:
$\mathbf{P}(1,1)$, $\mathbf{P}(1,2)$ % Matrix elements used to construct rotation
Output:
$\Theta_1(n)$ % 3 3 Givens rotation matrix
if $\mathbf{P}(1,2) = 0$ **then**
$\varpi_1 = 1$; $\varpi_2 = 0$ % No rotation needed
elseif $\mathbf{P}(1,1) = 0$ **then**
$\varpi_1 = 0$; $\varpi_2 = 1$ % Full rotation
else
$\theta_p = \mathbf{P}(1,2)/\mathbf{P}(1,1)$ % Tangent of rotation angle
$\varpi_1 = 1/\sqrt{1+\theta_p^2}$ % Cosine component
$\varpi_2 = \theta_p \cdot \varpi_1$ % Sine component
end
Construct output rotation matrix:

$$\Theta_1(n) = \begin{bmatrix} \varpi_1 & \varpi_2 & 0 \\ \mathrm{conj}(\varpi_2) & -\varpi_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Algorithm 5 HyperbolicRot.

Input:
$\mathbf{P}_1(1,1)$, $\mathbf{P}_1(1,3)$ % Elements for computing hyperbolic rotation
Output:
$\Theta_2(n)$ % 3 3 hyperbolic rotation matrix
if $\mathbf{P}_1(1,3) = 0$ **then**
$\varpi_3 = 1$, $\varpi_4 = 0$ % No rotation needed
elseif $|\mathbf{P}_1(1,1)| > |\mathbf{P}_1(1,3)|$ **then**
$\theta_p = \mathbf{P}_1(1,3)/\mathbf{P}_1(1,1)$ % Ratio of off-diagonal to diagonal element
else
$\theta_p = [\mathbf{P}_1(1,1)/\mathbf{P}_1(1,3)]^T$ % Transposed when denominator is larger
end
$\varpi_3 = 1/\sqrt{1-\theta_p^2}$ % Hyperbolic cosine
$\varpi_4 = \theta_p \cdot \varpi_3$ % Hyperbolic sine
Construct rotation matrix:

$$\Theta_2(n) = \begin{bmatrix} \varpi_3 & 0 & -\varpi_4 \\ 0 & 1 & 0 \\ \mathrm{conj}(\varpi_4) & 0 & \varpi_3 \end{bmatrix}$$

TABLE II
PARAMETER SETTINGS OF LMS, RLS, DR-RLS, PRLS, AND FARLS ALGORITHMS

parameter \ algorithms	LMS	RLS	DR-RLS	PRLS	FARLS
λ	\	0.98	0.999	0.98	0.999
p	9×10^{-3}	\	\	\	\
c	\	\	4	\	\
ϵ	\	\	\	10^{-4}	\
μ	\	\	\	$\lfloor L/2 \rfloor - 1$	\
ϑ	\	\	\	-0.5	\
η	\	\	\	\	1

with its counterparts. In addition, another test of communication simulations is designed to compare the BER of FAE with other adaptive equalizers. The algorithm parameters used in Sections IV-A and IV-B are shown in Table II, where p denotes the step size of the LMS algorithm and $\lfloor x \rfloor$ represents rounding down x. Each algorithm is independently run 5×10^2 times with the same data length to compute the average mean square deviation (MSD).

TABLE III
COMPUTATIONAL COMPLEXITY COMPARISON OF LMS, RLS, DR-RLS, PRLS, AND FARLS ALGORITHMS

algorithms	LMS	RLS	DR-RLS	PRLS	FARLS
computational complexity	$\mathcal{O}(L)$	$\mathcal{O}(L^2)$	$\mathcal{O}(L^2)$	$\mathcal{O}(L^2)$	$\mathcal{O}(L)$

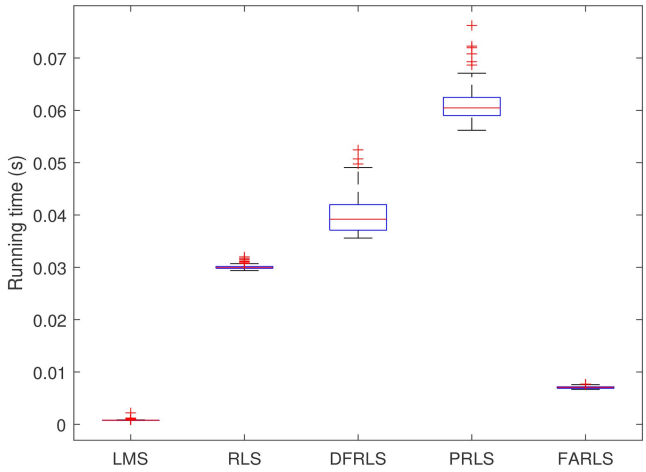


Fig. 5. Comparison of running time for LMS, RLS, DR-RLS, PRLS, and FARLS algorithms.

The definition of MSD is as follows:

$$\mathrm{MSD} = \frac{\|\mathbf{w} - \mathbf{h}\|_2}{\|\mathbf{h}\|_2} \tag{32}$$

where $\mathbf{h}$ is the vector of ideal filter weight coefficients.

A. Test of Algorithm Complexity

In this section, we measured the runtime of different algorithms with $N = 10^3$ and $L = 90$, using the runtime as a rough estimate of computational complexity. Table III is the complexity of different algorithms. The linear complexities of the LMS and FARLS algorithms are approximately equal, while that of the RLS, DR-RLS, and PRLS algorithms is one order of magnitude higher.

The results are shown in Fig. 5. It is evident that, in the RLS-based algorithms, the FARLS algorithm exhibits the shortest runtime. This aligns with the analysis of computational complexity, where the linear complexity of the FARLS algorithm is at the same order as the LMS algorithm, i.e., $\mathcal{O}(L)$.

B. Test of Learning Curve

In this section, we compare the convergence speed and steady-state MSD of the LMS, RLS, DR-RLS, PRLS, and FARLS algorithms. The filter length is variable in the tests and signal-to-noise ratio (SNR) is constant. The SNR is defined as

$$\mathrm{RSNR} = 10 \log_{10} \frac{p_t}{p_n} \tag{33}$$

where p_t is the power of the received signal and p_n denotes the noise power.

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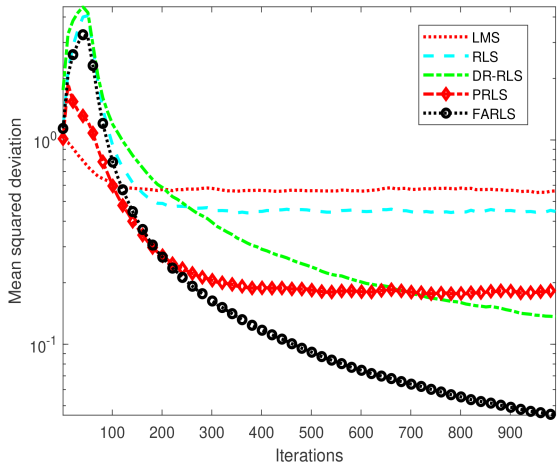


Fig. 6. RSNR = 0 dB, $L = 40$. The comparisons of learning curves for LMS, RLS, DR-RLS, PRLS, and FARLS algorithms.

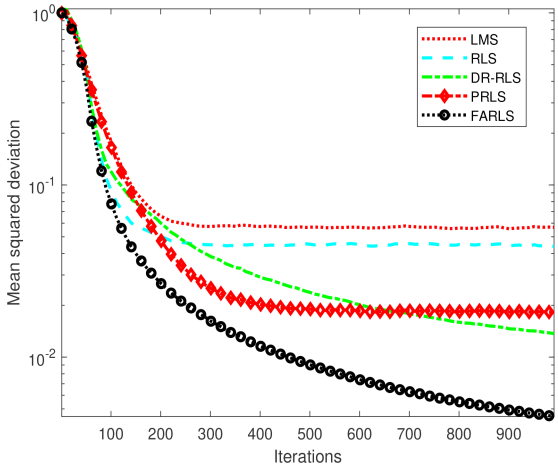


Fig. 7. RSNR = 10 dB, $L = 40$. The comparisons of learning curves for LMS, RLS, DR-RLS, PRLS, and FARLS algorithms.

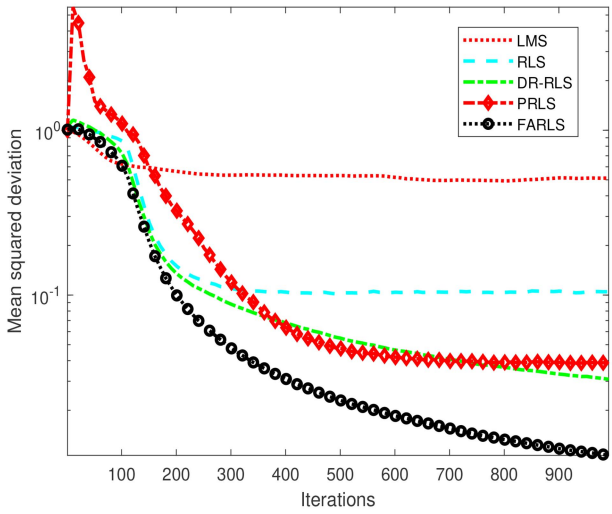


Fig. 8. RSNR = 10 dB and $L = 90$. The comparisons of learning curves using different algorithms.

TABLE IV
COMMUNICATION SYSTEM PARAMETERS SETTINGS IN SIMULATIONS

The environmental parameters	Value
Sampling frequency	96 kHz
Central frequency	15 kHz
Band width	10 kHz
roll-off factor	0.25
L	30
f_d	4 Hz
t_g	1s
t_u	1s
t_d	1s
K_{f1}	10^{-2}
K_{f1}	10^{-3}
N_t	$10^2/3 \times 10^2$ symbols
N_r	$N_r = 10N_t/N_r = 4N_t$

The simulation results are shown in Figs. 6–8. For $L = 40$, when RSNR = 0 dB, the results are shown in Fig. 6, LMS algorithm quickly converges to steady state, while DR-RLS and FARLS algorithms continue to exhibit a decreasing trend. Although DR-RLS and FARLS algorithms maintain a comparable rate of descent, the FARLS algorithm ultimately achieves the lowest MSD. The FARLS algorithm still provides the lowest steady-state error. For RSNR = 10 dB, the results are shown in Fig. 7, as the number of iterations reaches 3×10^2, LMS, RLS, and PRLS algorithms have reached steady-state, while DR-RLS and FARLS algorithms continue to exhibit a decreasing trend until reaching 10^3 iterations. Although DR-RLS and FARLS algorithms maintain a comparable rate of descent, the FARLS algorithm ultimately achieves the lowest MSD. When $L = 90$, it is evident that a similar scenario is observed in Fig. 8 as well. The FARLS algorithm still provides the lowest steady-state error.

C. UWA Communication Simulations

In this section, we conduct communication simulations and designed various training symbol lengths and modulation schemes to compare the BER performance of FAE with LMS, RLS, PRLS, and DR-RLS adaptive equalizers. Based on hardware parameters and practical limitations, we set the communication system parameters as shown in Table IV. The algorithm parameters used in the simulations are shown in Table II, with the slight difference that $p = 10^{-2}$. The transmitted information consists of 262 145 bits, which are divided into data frames as illustrated in Fig 1 and transmitted and received according to the process outlined in Fig 2. Each frame begins with the up-LFM signal, and the final frame ends with the down-LFM signal.

We introduce the BER as the evaluation criterion for equalization performance. The definition of BER is

$$\text{BER} = \frac{n_e}{n_t} \tag{34}$$

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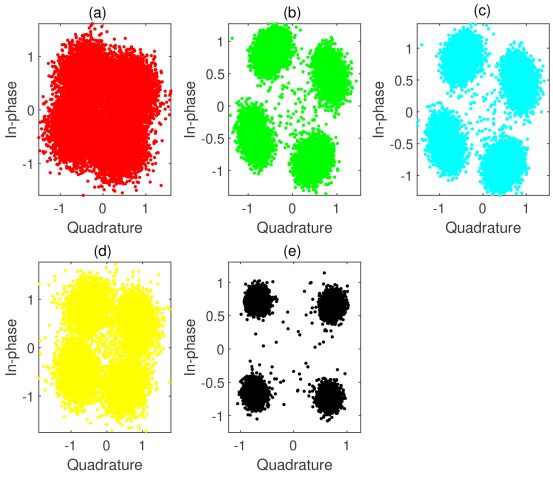


Fig. 9. 4PSK modulation, $N_t = 10^2$ symbols, $N_s = 10N_t$. The comparisons of the constellation diagrams obtained in simulations LMS, RLS, DR-RLS, PRLS, and FAE. (a) LMS. (b) RLS. (c) DR-RLS. (d) PRLS. (e) FARLS.

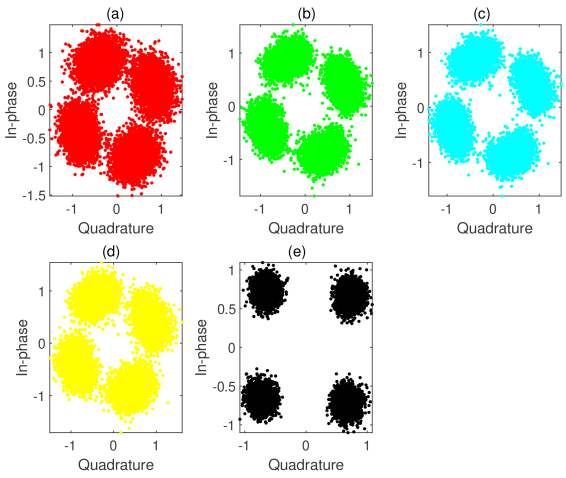


Fig. 10. 4PSK modulation, $N_t = 3 \times 10^2$ symbols, $N_s = 4N_t$. The comparisons of the constellation diagrams obtained in simulations using different adaptive equalizers. (a) LMS. (b) RLS. (c) DR-RLS. (d) PRLS. (e) FARLS.

where n_e and n_t, respectively, represent the total number of erroneous bits at the receiver and the total number of transmitted bits.

Fig. 9–11 are the simulation results for 4PSK modulation. When $N_t = 10^2$ symbols and $Ns = 10N_t$, the LMS, RLS, DR-RLS, and PRLS adaptive equalizers exhibit similar characteristics in constellation Fig. 9, with a large number of constellation points failing to converge together. While the FAE provides better compensation for phase rotation, resulting in the corresponding image with the lowest BER. From constellation Fig. 10, it is evident that increasing N_t to 3×10^2 symbols is beneficial, especially for the LMS adaptive equalizer. But the FAE still maintains optimal accuracy.

The above conclusions are further confirmed in Fig. 11, which shows the BER for the five adaptive equalizers under different N_t lengths. Different colors represent different ranges of BER, where yellow represents the most severe BER, while

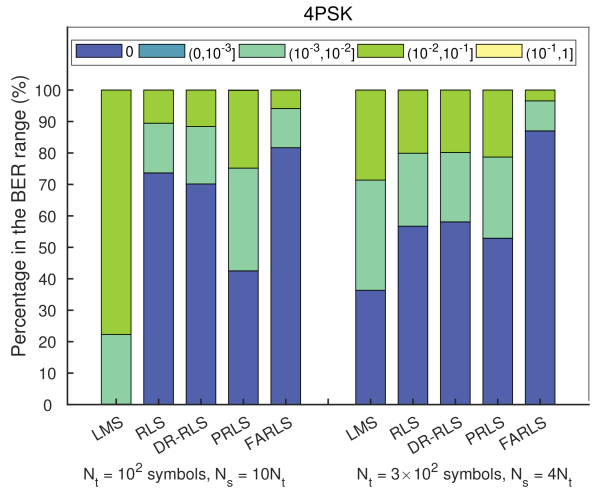


Fig. 11. BER comparisons of the figures in simulations among LMS, RLS, DR-RLS, PLRS, and FAE.

TABLE V
BER COMPARISONS OF LMS, RLS, DR-RLS, PRLS, AND FARLS ADAPTIVE EQUALIZERS IN LOW SNR CONDITIONS WITH AT-SEA EXPERIMENTS

$\frac{E_b}{N_0}$ (dB) \ Alg.	LMS	RLS	DRRLS	PRLS	FARLS
4	0.0648	0.0511	0.0426	0.0402	0.0293
3	0.0655	0.0524	0.0438	0.0435	0.0303
2	0.0696	0.0556	0.0542	0.0545	0.0305
1	0.0707	0.0579	0.0559	0.0566	0.0339

blue represents the BER = 0. It is evident that as the length of N_t increases, the package of the LMS adaptive equalizer with BER = 0 increases from 0% to 36%, consistent with the description given above. The BER of the RLS-based adaptive equalizers is lower. Regardless of the value of N_t, the packages with the BER $< 10^{-3}$ always account for more than 70%. Among RLS-based adaptive equalizers, the FAE consistently exhibits the best BER performance, with packages achieving the BER $< 10^{-3}$ exceeding 90%.

To evaluate the BER performance of different equalizer algorithms at various low E_b/N_0 levels, We then added different levels of additive white Gaussian noise to simulate the required E_b/N_0 conditions. The corresponding BER results for each E_b/N_0 level are presented in Table V. As shown, our FARLS adaptive equalizer consistently achieves lower BER than the LMS, RLS, DR-RLS, and PRLS methods.

For 8PSK modulation, the constellation diagrams driven by different adaptive equalizers are shown in Figs. 12 and 13. The case of $N_t = 10^2$ symbols corresponds to Fig. 12. One can see that the FAE exhibits superior noise resilience compared to the other equalizers in Fig. 12. Even after eliminating the influence of insufficient training sequence length, i.e., with $N_t = 3 \times 10^2$ symbols, the BER performance of the LMS adaptive equalizer still lags behind that of the RLS-based adaptive equalizers, especially the FAE. The BER results showed in Fig. 14 also stands for that the FAE exhibits the best accuracy under 8PSK

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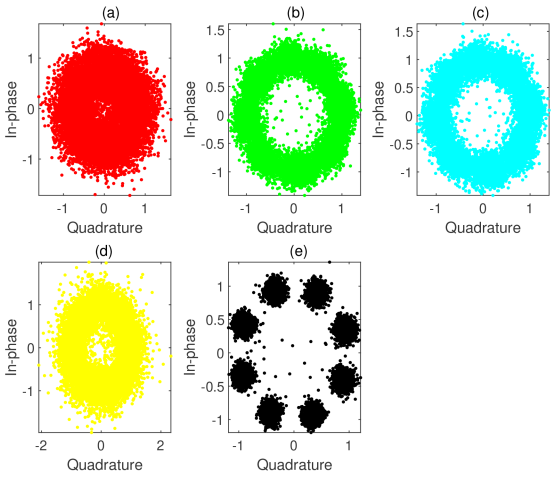


Fig. 12. 8PSK modulation, $N_t = 10^2$ symbols, $N_s = 10N_t$. The comparisons of the constellation diagrams obtained in simulations using different equalizers. (a) LMS. (b) RLS. (c) DR-RLS. (d) PRLS. (e) FARLS.

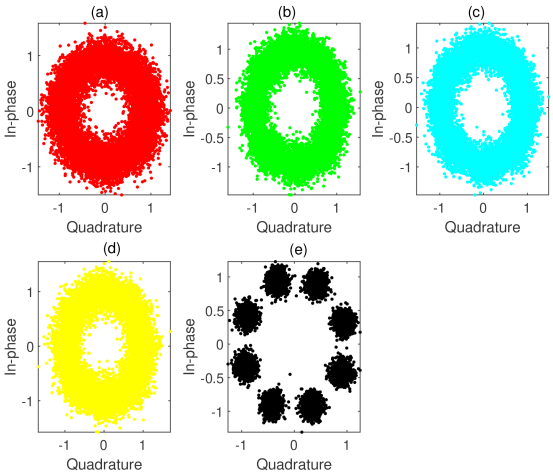


Fig. 13. 8PSK modulation, $N_t = 3 \times 10^2$ symbols, $N_s = 4N_t$. The comparisons of the images obtained in simulations for FAE and its counterparts. (a) LMS. (b) RLS. (c) DR-RLS. (d) PRLS. (e) FARLS.

modulation. Specifically, the percentage of packages with BER $< 10^{-3}$ remains above 80% for the FAE, far exceeding the combined percentage of blocks with BER $< 10^{-3}$ for other adaptive equalizers.

The simulations have already verified that the accuracy of the LMS adaptive equalizer decreases when the $N_t = 10^2$ symbols. Therefore, in UWA experiments, to ensure fair performance comparison among all equalizers, we set $N_t = 3 \times 10^2$ symbols and $N_s = 4N_t$.

V. At-Sea Experiment

In this section, Section V-A is the analysis of the BER performance of FAE based on real data, while Section V-B explores the potential development directions of FAE and FARLS.

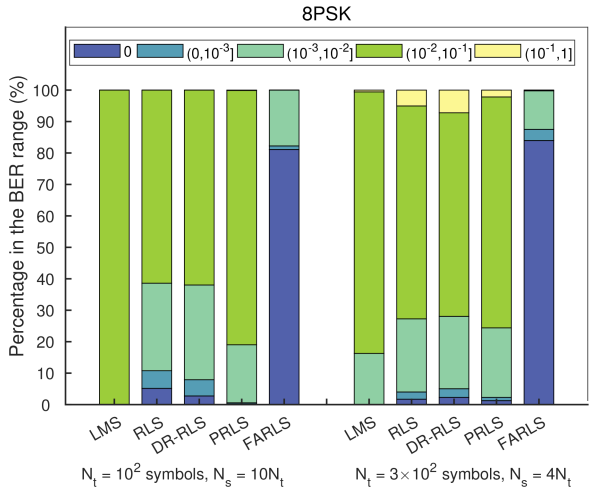


Fig. 14. BER comparisons of the figures in simulations among LMS, RLS, DR-RLS, PRLS, and FAE.

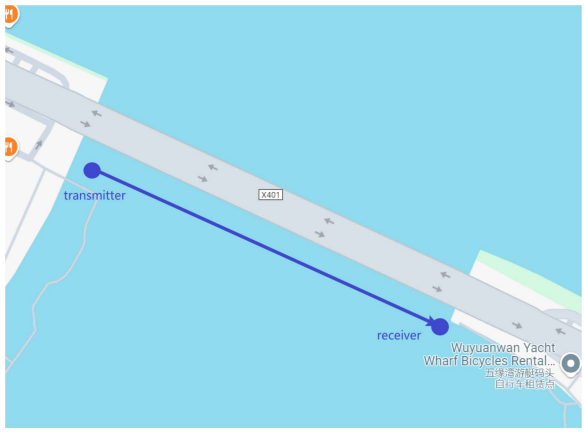


Fig. 15. At-sea experiments site.

A. Real Data Analysis

To verify the performance in combating Doppler shift and robustness to noise of the proposed FAE, we conduct at-sea experiments. The UWA experiments take place on the morning of 17 January, 2024, in Wuyuan Bay, Xiamen, China, with the specific site shown in Fig. 15. The average depth of Wuyuan Bay is 5–6 m. Both the transmitter and receiver are located underwater at a depth of 3 m, and the transmission distance is set to 700 m. The sound speed profile is shown in Fig. 16. One can see that a slight negative gradient. Fig. 17 is the channel impulse response, where the horizontal and vertical axis are the delay time and real time, respectively.

The major noise interference in the at-sea experiments come from the surrounding urban environmental noise, propeller noise, and marine biological noise. The experimental site is near an overpass and an underwater tunnel, where industrial activities generated significant noise. The working propellers of nearby boats and the activity of marine organisms also pose challenges to the quality of underwater data transmission. In addition, the relative motion between the receiving and transmitting vessels

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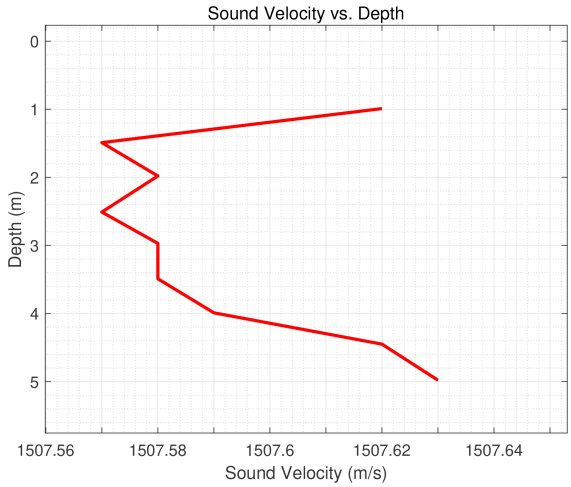


Fig. 16. Underwater sound speed profile in Wuyuan Bay.

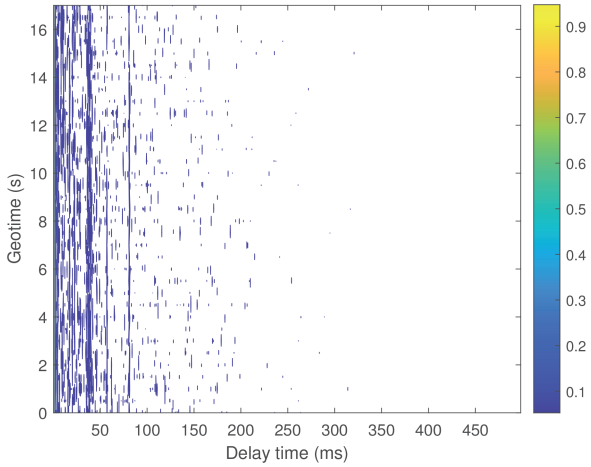


Fig. 17. Channel impulse response of the at-sea experiments.

causes Doppler shifts, making the design of an efficient and reliable receiver at the receiving end essential for reconstructing the data.

In the experiments, the algorithm and communication system parameters are specified in Tables II and VI, respectively. The size of the transmitted image is 103×103 and converted into 42 437 bits. Initially, interleaving is applied, and the subsequent steps for constructing the transmission frame are the same as in the simulation. Fig. 18 is the short-time Fourier transform spectrogram of the transmitted data, revealing the inclusion of four data frames. The modulation scheme of the transmitted signal is 4PSK.

The RSNR is approximately 12 dB. The reconstructed images and their corresponding constellation figures are shown in Figs. 19 and 20. The images processed using FAE exhibit the most robust communication quality, with superior noise robustness and phase drift resistance compared to other equalizers. Fig. 21 is the BER comparison among the five equalizers. The BER performance of RLS-based adaptive equalizers surpasses that of LMS adaptive equalizer, with FAE exhibiting a BER

TABLE VI
COMMUNICATION SYSTEM PARAMETER SETTINGS IN EXPERIMENTS

The environmental parameters	Value
Sampling frequency	96 kHz
Central frequency	10.25 kHz
Band width	3.5 kHz
roll-off factor	0.1667
symbol rate	3 kBaud/s
t_g	0.05 s
t_u	0.1 s
t_d	0.1 s
K_{f1}	10^{-2}
K_{f1}	10^{-3}
N_t	3×10^2 symbols
N_r	1.2×10^3 symbols

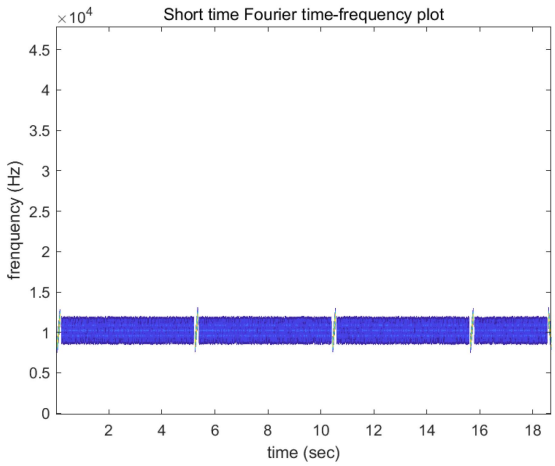


Fig. 18. Short-time fourier transform spectrogram of the transmitted data.

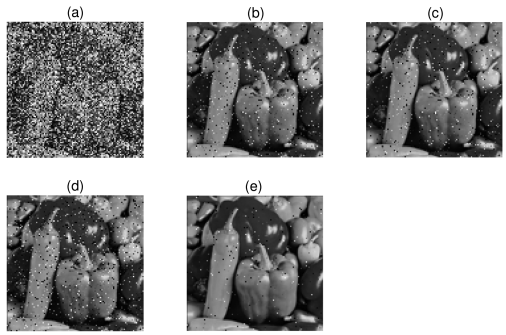


Fig. 19. Reconstructed image comparisons of the FAE and its counterparts. (a) LMS. (b) RLS. (c) DR-RLS. (d) PRLS. (e) FARLS.

percentage in $(10^{-3}, 10^{-2}]$ well above 90%, far exceeding its counterparts.

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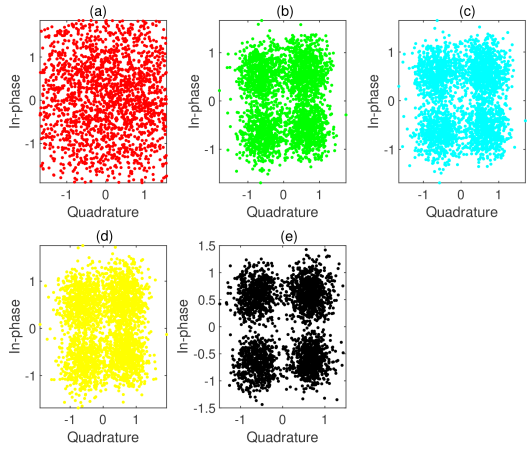


Fig. 20. Constellation Comparisons of the FAE and its counterparts. (a) LMS. (b) RLS. (c) DR-RLS. (d) PRLS. (e) FARLS.

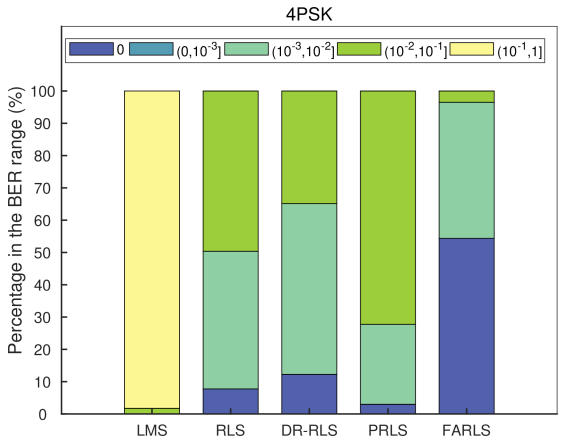


Fig. 21. BER comparisons of LMS, RLS, DR-RLS, PRLS, and FAE.

B. Future Directions

1) Restating the Practical Implications: The FAE, with its linear complexity and ability to rapidly compensate for the rotation of channel taps, significantly improves the efficiency and accuracy of underwater channel equalization. This capability is particularly valuable in highly dynamic environments, such as in ocean exploration, uncrewed underwater vehicle communication, and subsea monitoring systems, where it effectively addresses signal distortion and phase drift caused by Doppler shift and complex UWA conditions.

2) Potential Impact on UWA Communication: The FAE's ability to reduce BER and enhance data transmission reliability has profound implications for extending the effective range of communication links, increasing transmission rates, and maintaining communication stability in noisy environments. These features make the FAE highly promising for practical deployment, especially in scenarios requiring high precision and real-time processing in UWA communication systems.

3) Future Research Directions and Applications: Exploring the integration of the FARLS algorithm with machine learning techniques to adaptively adjust algorithm parameters, thereby improving channel equalization outcomes. The design principles and technical advantages of the FARLS algorithm could also be applied to other fields, such as seismic signal processing and medical ultrasound imaging, thus broadening its application scope.

VI. Conclusion

In order to effectively combat ISI in UWA communication, we propose a low-complexity equalization scheme for phase coherent systems. We perform a structural transformation on the input vector of the equalizer and introduce the transformed structure into an array form. The FARLS algorithm with linear $\mathcal{O}(L)$ complexity is derived and compared with LMS, RLS, DR-RLS, and PRLS algorithms, considering both computational complexity and convergence speed. The test results indicate that the FARLS algorithm has advantages in both aspects. Furthermore, we provide the process of phase coherent UWA communication and design data frames with LFM signals and guard space. Detailed algorithm parameters and environmental parameters are provided to validate the feasibility of the proposed FARLS adaptive equalizer in UWA communication simulations and at-sea experiments. The results of the at-sea experiments indicate that, under comparable computational complexity, the percentage of blocks with BER $>10^{-2}$ for the FAE is less than 5%, significantly lower than its counterparts. The FAE demonstrates better noise robustness and resistance to Doppler shift.

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